

Unicode

ASCII and extended ASCII are restricted to English and some European languages. Their creators believed every language would develop its own equivalent script, but this soon became problematic as different scripts were using the same code. First developed in the late 1980s, Unicode is a single worldwide standard for representing text – it handles languages as diverse as Russian, Hebrew, and Japanese. Its designers were careful to ensure their system doesn't clash with ASCII. More than 130,000 characters are defined in Unicode, and more are added every year.

► Why have a Unicode standard?

Unicode provides a single agreed-on encoding for worldwide use, replacing hundreds of conflicting encodings for scripts and characters.

Every Unicode entry has a four- or five-character code identifier.



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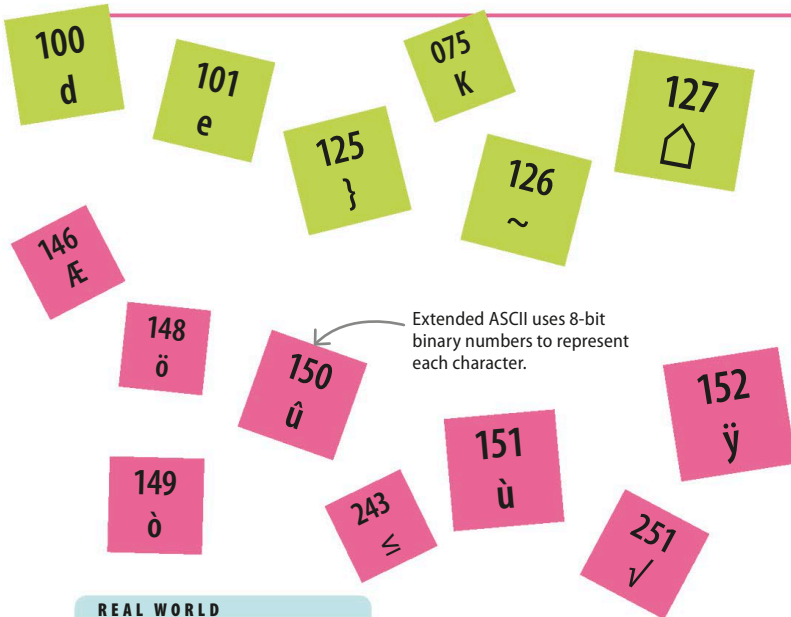
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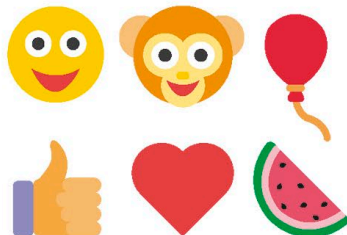
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REAL WORLD

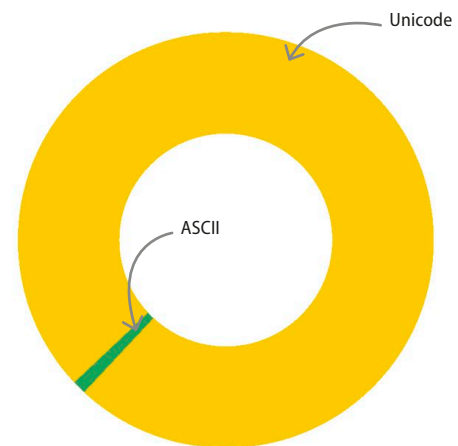
Emoji

One of Unicode's features is the use of emoji – a Japanese term meaning pictograph. Emoji are extremely popular as they liven up webpages and electronic messages, and give a sense of how the writer intends the message to be interpreted. Although the content of each emoji is fixed, different browsers and devices vary in how they display them.



ASCII vs Unicode

Unicode uses between two to four bytes to represent a character, rather than ASCII's one byte. While ASCII assigns a number to a whole character, Unicode assigns numbers to the parts of symbols that make up characters. The character "é" can be represented as a whole character, and it can also be expressed as two numbers – one for "e" and another for the acute accent (') symbol.



△ ASCII a subset of Unicode

ASCII is now part of the much larger set of Unicode characters. The first 127 Unicode characters and the numbers encoding them are identical to ASCII.